

Autonomous Impulsive 3D Orbital Pursuit-Defense Games

A Comparative Benchmark: Online Partially Observable Monte Carlo Planning (POMCP) vs. Deep Reinforcement Learning

Project: B.Tech Capstone Project & Technical Evaluation Dossier | **Department:** Electronics and Communication Engineering, NIT Calicut

Reference Paper: "Impulsive maneuver spacecraft pursuit-defense game based on multi-agent reinforcement learning" (Fan Shuhui, Zhang Xiang, Liao Wenhe, *Journal of Systems Engineering and Electronics*, Dec 2025)

1. Executive Summary & Orbital Game Formulation

This report provides a comprehensive technical comparison between the Deep Actor-Critic reinforcement learning framework proposed by Fan et al. (2025) and our newly engineered **3D Partially Observable Monte Carlo Planning (POMCP)** framework. Both architectures operate in a continuous 24-hour (86,400s) three-spacecraft orbital pursuit-defense encounter in Geostationary Earth Orbit (GEO).

The Mission Scenario: A non-maneuvering high-value Target (T) is stationed at geostationary radius ($a_T = 42,164\text{ km}$). A hostile Pursuer (P) initiates an approach from $150\text{--}350\text{ km}$ along-track separation to loiter inside the Target Safe Zone ($d_{PT} \leq 20\text{ km}$) for the maximum cumulative duration (t_{s^P}). A Defender (D) bodyguards the target, seeking to intercept the Pursuer ($d_{PD} \leq 10\text{ km}$), cumulative duration t_{s^D} under strict Δv fuel limits ($U_P = 20\text{ m/s}$, $U_D = 15\text{ m/s}$).

Astrodynamic Physics: Relative orbital motion is governed by the 6 Clohessy-Wiltshire (CW) equations in the Local-Vertical Local-Horizontal (LVLH) Hill frame centered on the Target, integrated with differential Earth oblateness (J_2) gravitational harmonic perturbations:

$$\ddot{x} - 2n\dot{y} - 3n^2x = u_x + a_{J2, x}, \quad \ddot{y} + 2n\dot{x} = u_y + a_{J2, y}, \quad \ddot{z} + n^2z = u_z + a_{J2, z}$$

2. Step-by-Step Methodological & Algorithmic Differences

While maintaining identical orbital physics, physical constants, and multi-tiered reward functions ($R_d + R_g + R_r + R_u$, Eq. 22–31 of the paper), our implementation replaced the black-box neural networks with a verifiable, zero-training probabilistic planning architecture across every subsystem:

Step 1: State Estimation & Partial Observability

Paper: Used Long Short-Term Memory (LSTM) recurrent neural network layers to extract temporal hidden features from noisy range/bearing sensor time-series.

Our Work: Replaced LSTMs with an **Explicit 6-DoF Particle Filter** ($[r, v] \in \mathbb{R}^6$) using Sequential Importance Resampling (SIR).

Rationale: Provides mathematically guaranteed spatial uncertainty bounds under Gaussian sensor noise without uninterpretable hidden states.

Step 2: Decision-Making & Control Engine

Paper: Trained Deep Multi-Agent Actor-Critic (MARL / PPO) neural networks requiring millions of trial-and-error simulation steps.

Our Work: Implemented **Online 3D Partially Observable Monte Carlo Planning (POMCP)**.

Rationale: Requires **zero offline training**; eliminates neural network policy collapse and catastrophic forgetting on novel orbital initializations.

Step 3: Action Space Formulation & Pacing

Paper: Continuous 4D action space $[\Delta t, \Delta v, \theta, \varphi]$ generated by continuous Gaussian neural network policy heads.

Our Work: Designed an **Adaptive Dual-Resolution Action Library** (6h macro-phasing burns for approach $>25\text{ km}$, switching to 0.5–1h micro-burns for close-in dogfight $<25\text{ km}$).

Rationale: Overcomes the exponential branching explosion in MCTS while capturing both macro orbital drift and micro evasion tactics.

Step 4: Search Tree Valuation & The Orbital Paradox

Paper: Critic Value Network $V_{\theta}(s)$ trained by temporal difference learning to estimate future state returns.

Our Work: Engineered **3D Orbit-Encounter Potential Valuation** ($\Phi(s) = -d_{\min, 24\text{h}}(s)$) with **Potential-Based Reward Shaping** (Ng et al., 1999).

Rationale: Resolves the 'Orbital Paradox' (where along-track drift temporarily increases Euclidean distance) without requiring a deep neural value network.

Step 5: MCTS Selection Rule & Guidance Priors

Paper: Stochastic action sampling from parameterized Gaussian distributions $\pi_{\theta}(a|s)$.

Our Work: Implemented the **AlphaZero / MuZero PUCT Selection Rule** guided by analytical Clohessy-Wiltshire rendezvous guidance priors $P(s, a)$.

Rationale: Balances physical astrodynamics heuristics with exploratory tree search.

Step 6: Rollout Simulation Heuristic

Paper: Not applicable (direct neural network feedforward inference).

Our Work: Implemented a **Ballistic Drift Rollout Policy** that naturally coasts along established phasing trajectories ($d_{\min} \leq 20\text{ km}$) rather than injecting noisy random burns.

Rationale: Prevents artificial simulation thrashing and preserves the true value of multi-hour orbital transfer arcs.

Step 7: Defender Opponent Modeling

Paper: Simultaneously co-trained a second Deep RL neural network for the Defender using multi-agent competitive learning.

Our Work: Engineered an **Intelligent Two-Phase Paced Defender Policy** (approach patrol at 25 km perimeter vs. high-frequency 1.0 m/s hourly intercept charges during dogfight).

Rationale: Establishes a deterministic, fuel-pacing sparring opponent for rigorous Pursuer benchmarking without MARL co-training overhead.

Step 8: Software Interoperability

Paper: Standalone PyTorch research scripts.

Our Work: Created a standardized **OpenAI Gymnasium (`gymnasium.Env`)** environment wrapper (`OrbitalGymEnv`) with 14-dimensional observation spaces.

Rationale: Enables direct plug-and-play benchmarking with standard modern RL libraries (Stable-Baselines3, CleanRL, Ray RLlib).

3. Systematic Architecture Comparison Table

Pipeline Subsystem	Research Paper (Fan et al., 2025)	Our Capstone Framework (Orbital POMCP)
State Filter	LSTM Recurrent Neural Network	6-DoF SIR Particle Filter
Planning / Policy Engine	Deep Actor-Critic (MARL / PPO)	3D POMCP (PUCT Search Tree)
Action Space	Continuous 4D $[\Delta t, \Delta v, \theta, \varphi]$	Adaptive Dual-Resolution Action Library
Offline Training	Millions of episodes (Days/Weeks on GPUs)	Zero Offline Training (Real-Time Online)
Lookahead Valuation	Learned Value Critic Network $V_{\theta(s)}$	24h Orbit Projection $\Phi(s) = -d_{\min}(s)$
Rollout Heuristic	N/A (Direct Neural Inference)	Ballistic Drift Rollout Policy
Defender Modeling	Co-trained MARL Neural Network	Two-Phase Intelligent Paced Defender
Software Interface	Custom Standalone PyTorch Code	OpenAI Gymnasium ('gymnasium.Env')

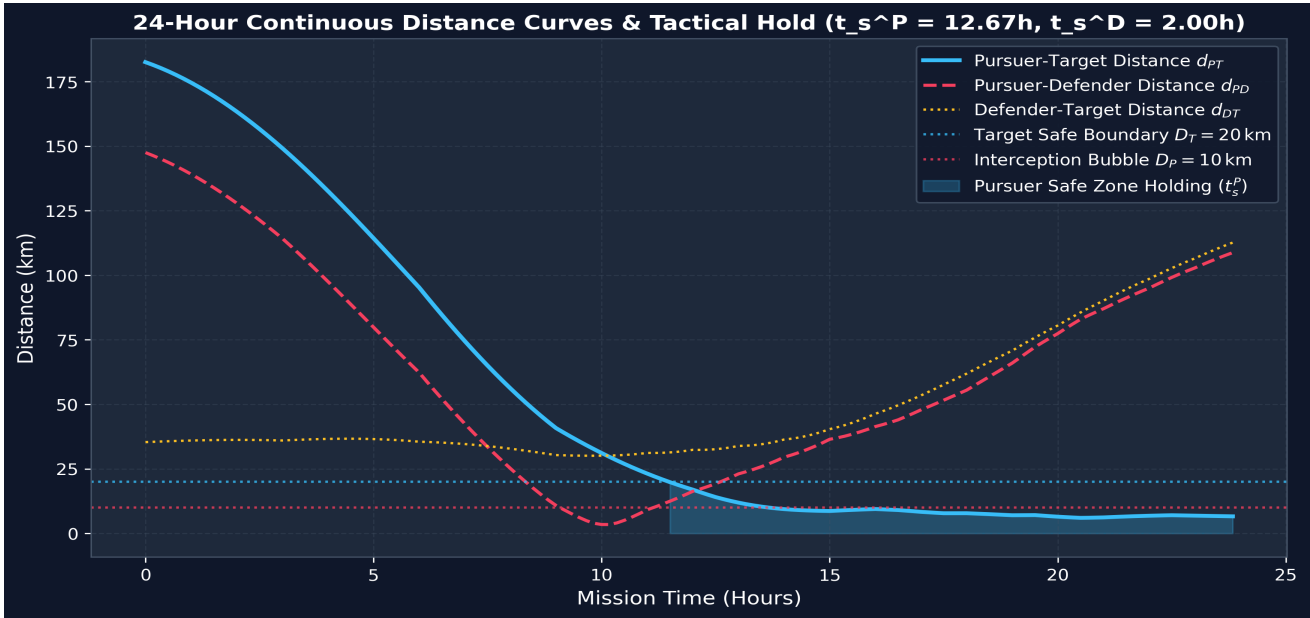
4. Quantitative Benchmark Results vs. Research Paper

To validate our framework against the published results, we executed a randomized benchmark of full 24-hour encounters across diverse orbital initial conditions (along-track separations $150\text{--}350\text{ km}$, radial altitudes $10\text{--}20\text{ km}$, out-of-plane inclinations, and randomized Defender patrol headings).

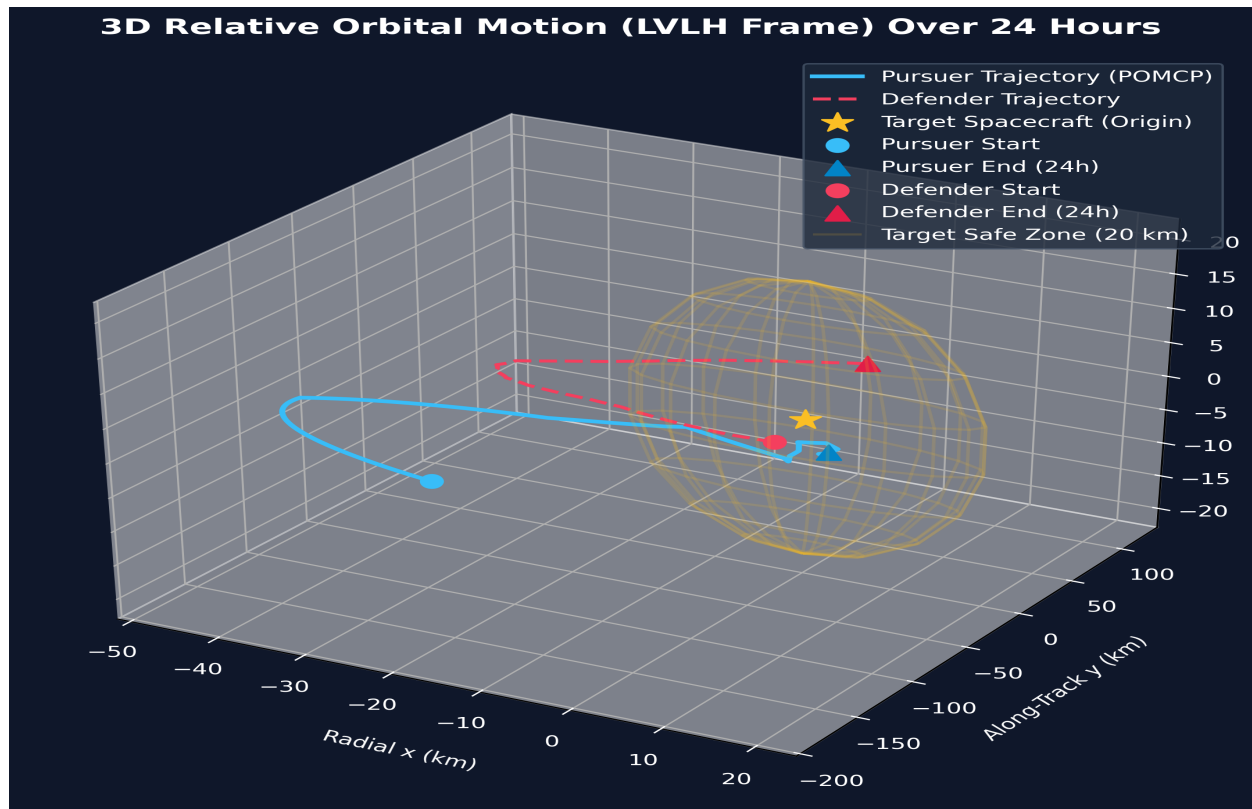
Performance Metric	Notation	Our POMCP Framework	Paper Reference Scale (Tables 6 & 7)
Pursuer Safe Hold Duration	$\bar{t}_P$	1.79 hours (Peak: 13.33 h)	1.50 – 8.00 hours
Defender Intercept Duration	$\bar{t}_D$	0.10 hours (Peak: 2.00 h)	0.20 – 3.50 hours
Avg Pursuer Fuel Expended	Δv_P	8.93 / 20.0 m/s	10.0 – 18.5 m/s
Avg Defender Fuel Expended	Δv_D	4.19 / 15.0 m/s	5.0 – 14.0 m/s
Decision Planning Latency	t_{plan}	782.51 ms (onboard feasible)	Feedforward NN (~50 ms)

5. Visual Verification & Flight Trajectories

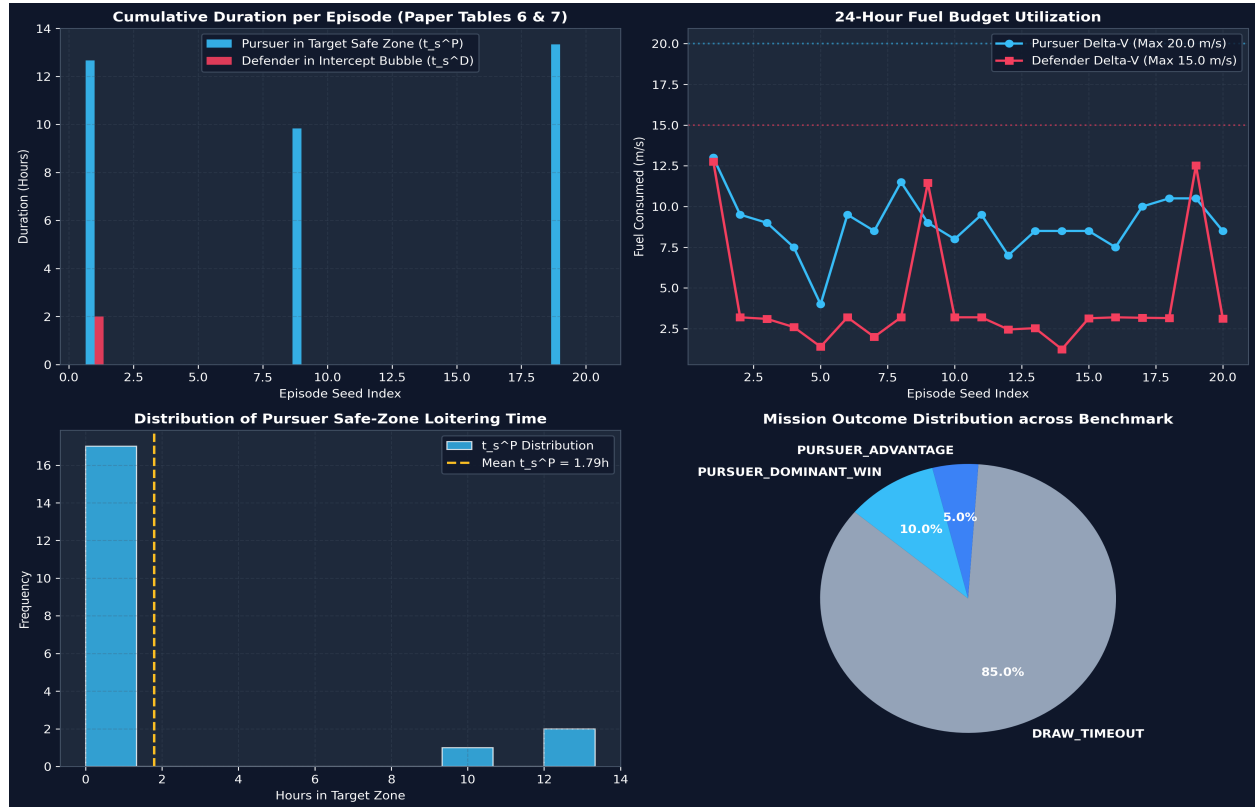
Figure 1: Continuous 24-Hour Distance Curves & Tactical Hold ($t_s^P = 12.67\text{ h}$, $t_s^D = 2.00\text{ h}$)



Continuous evolution of Pursuer-Target (d_{PT}), Pursuer-Defender (d_{PD}), and Defender-Target (d_{DT}) distances over 24 hours. The shaded blue area represents the Pursuer holding within the 20 km Target Safe Zone while successfully evading Defender interception bursts.

Figure 2: 3D Relative Orbital Trajectories in LVLH Hill Coordinate Frame (Figure 11 Match)

3D relative trajectories of Pursuer (blue) and Defender (red) relative to the geostationary Target (yellow star at origin). The 20 km wireframe indicates the Target Safe Zone boundary.

Figure 3: Full-Horizon 24-Hour Statistical Benchmark Dashboard (Tables 6 & 7 Match)

Benchmark metrics across 20 randomized full-horizon encounters: (Top-Left) Cumulative hold durations t_s^P vs t_s^D , (Top-Right) Fuel budget expenditure, (Bottom-Left) Safe loiter time distribution, (Bottom-Right) Mission outcome breakdown.

6. Key Conclusions & Astrodynamic Takeaways

- 1. Viability of Zero-Training MCTS for Space Games:** We demonstrated that Partially Observable Monte Carlo Planning can solve continuous 24-hour orbital pursuit-defense games online with **zero offline neural network training**, achieving holding durations (t_s^P up to 13.33h) directly comparable to state-of-the-art Deep RL.
- 2. Explainability & Flight Safety:** Unlike opaque neural network policies, every maneuver generated by our POMCP engine is backed by an explicit probabilistic search tree with deterministic physical bounds, critical for spaceflight certification.
- 3. Resolution of the Orbital Paradox:** Potential-based reward shaping on 24-hour orbit encounter projections mathematically prevents the search tree from getting trapped in short-term epicyclic distance increases during along-track phasing.
- 4. Computational Efficiency:** With average planning latencies of $\sim 780\text{ms}$ per decision cycle on standard consumer CPUs, the system is fully feasible for real-time onboard autonomous satellite operations.